

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Haashini P (192524047) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total

transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate	Good understanding with minor gaps	Basic understanding with some	Poor understanding; major misconceptions

		understandin g of concepts		misconcepti on s	
Explanat ion	25%	Detailed, well explained answers with rele vant examples	Clear explanation with adequate de tails	Limited explanation; lacks depth	Poor or unclear explanat ion
Applicat ion	20%	Effectivel y applies concepts with stro ng analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organiz ation	15%	Well structured answers with logi cal flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, in coherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = **1500 bytes**
- Transport layer segment size = **50 bytes**
- Data Link payload = **40 bytes**
- Header per frame = **40 bytes**

Solution:

Number of segments

$$= 1500 \div 50$$

$$= \mathbf{30 \text{ segments}}$$

Each segment (50 bytes) is divided into

$$50 \div 40 = 1.25$$

So, **2 frames** are required.

Total frames

$$= 30 \times 2$$

$$= \mathbf{60 \text{ frames}}$$

Total header overhead

$$= 60 \times 40$$

$$= \mathbf{2400 \text{ bytes}}$$

Ignoring propagation delay,

Transmission time depends only on total data and transmission rate.

Explanation:

- Transport Layer divides the file into segments.

- Data Link Layer converts each segment into frames by adding headers.
- Headers help in error detection and reliable communication.

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Window size = 6
- Frames = 48
- Frame size = 1024 bytes
- One frame in every window is lost.

Solution:

Number of transmission windows

$$= 48 \div 6$$

= 8 windows

One frame lost in every window

Retransmissions

= 8 frames

Explanation:

Sliding window allows multiple frames to be sent before waiting for acknowledgement. This increases speed and improves channel utilization.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20- byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- Packet size = **12000 bytes**
- MTU = **1500 bytes**
- Header = **20 bytes**

Solution:

Maximum data per fragment

$$= 1500 - 20$$

$$= \mathbf{1480 \text{ bytes}}$$

Number of fragments

$$12000 \div 1480 = 8.1$$

So,

$$= \mathbf{9 \text{ fragments}}$$

Header overhead

$$= 9 \times 20$$

$$= \mathbf{180 \text{ bytes}}$$

Explanation:

Network Layer breaks the packet into fragments so that each fragment fits inside the MTU. The destination system combines all fragments to recover the original packet.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Window size = **6**
- Frames = **48**
- One frame lost in every window.

Solution:

Transmission windows

$$= 48 \div 6$$

$$= \mathbf{8}$$

Retransmissions

= **8 frames**

Explanation:

Flow control prevents the receiver from getting overloaded and increases communication efficiency.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given:

- File = **600 MB**
- Checkpoint every **60 MB**
- Failure after **390 MB**

Solution:

Number of checkpoints

$$600 \div 60$$

= **10 checkpoints**

Completed checkpoints before failure

$$390 \div 60 = 6 \text{ remainder } 30$$

So,

Last checkpoint = **360 MB**

Data retransmitted

$$390 - 360$$

= **30 MB**

Explanation:

Session Layer checkpoints reduce retransmission because communication resumes from the last checkpoint instead of the beginning.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead

to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original file = **900 MB**
- Compression = **40%**
- Encryption = **5% of compressed size**

Solution:

Compressed size

$$= 900 \times 60\%$$

$$= \mathbf{540\ MB}$$

Encryption overhead

$$= 540 \times 5\%$$

$$= \mathbf{27\ MB}$$

Final transmitted size

$$= 540 + 27$$

$$= \mathbf{567\ MB}$$

Percentage reduction

$$= ((900 - 567) / 900) \times 100$$

$$= \mathbf{37\%}$$

Explanation:

Compression reduces transmission size while encryption protects data confidentiality.

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given:

- File = **3 GB**
- Throughput = **120 MB/s**
- Request size = **3 MB**

Solution:

Convert

3 GB = 3072 MB

Number of requests

$3072 \div 3$

= 1024 requests

Transmission time

$3072 \div 120$

= 25.6 seconds

Explanation:

Application Layer ensures reliable transfer using FTP by handling requests and responses.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given:

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data Link = 2 ms

Physical = 1 ms

Propagation = 18 ms

Transmission = 12 ms

Solution:

Total delay

$= 4 + 3 + 5 + 2 + 1 + 18 + 12$

= 45 ms

Explanation:

Each OSI layer introduces processing delay. Propagation and transmission delays occur during actual data transfer.

9.A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by

different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

- Data = **80 MB**
- Frame size = **1600 bytes**
- Overhead = **80 bytes**

Solution:

Useful data per frame

$$= 1600 - 80$$

$$= \mathbf{1520 \text{ bytes}}$$

80 MB

$$= 80 \times 1024 \times 1024$$

$$= \mathbf{83,886,080 \text{ bytes}}$$

Frames required

$$= 83,886,080 \div 1520$$

$$\approx \mathbf{55,188 \text{ frames}}$$

Total transmitted data

$$= 55,188 \times 1600$$

$$= \mathbf{88,300,800 \text{ bytes}}$$

Efficiency

$$= (1520/1600) \times 100$$

$$= \mathbf{95\%}$$

Explanation:

Protocol overhead decreases transmission efficiency because part of every frame is reserved for headers.

10. A hospital transfers a patient's medical record of 240 MB through an IoT enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across

the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given:

- Original record = **240 MB**
- Compression = **30%**
- Segment size = **1040 bytes**
- Header = **60 bytes**

Solution:

Compressed size

$$= 240 \times 70\%$$

$$= \mathbf{168 \text{ MB}}$$

Useful data per segment

$$= 1040 \text{ bytes}$$

168 MB

$$= 168 \times 1024 \times 1024$$

$$= \mathbf{176,160,768 \text{ bytes}}$$

Number of segments

$$= 176,160,768 \div 1040$$

$$\approx \mathbf{169,385 \text{ segments}}$$

Header overhead

$$= 169,385 \times 60$$

$$= \mathbf{10,163,100 \text{ bytes}}$$

Final transmitted data

$$= 176,160,768 + 10,163,100$$

$$= \mathbf{186,323,868 \text{ bytes}}$$

Explanation:

- Presentation Layer compresses the data.
- Transport Layer divides it into segments.
- Data Link Layer adds headers.
- Physical Layer transmits the data securely and reliably.